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Designing for Sovereignty

How digital architectures can support democratic governance

This brief memorandum explains in non-technical terms why certain digital architectures align better with democratic governance and free markets than others.

When a digital system is hard to replace (e.g. because of network effects or two-sided markets), has a great diversity of societal uses, and sets the rules for how its users can use it then we consider it to be **digital infrastructure**. Just as with physical infrastructure, digital infrastructure holds power because it is a critical input to economic production and structures social activity. There are *many* kinds of digital infrastructure: social media, search, browsers, operating systems, cloud, advertising networks, productivity systems, app stores, secure chats, AI components, standards, digital payments, identity systems, and more.

Sovereignty is the ability to make and enforce rules in one's own jurisdiction. When an infrastructure operator evades partly or fully jurisdictional oversight, the rules it sets to govern its users compete with and potentially supersede government-set laws. This happens for instance when businesses have to pay higher taxes to an app store than to the state, when social and search determine the rules of free speech and journalistic relevance, when ad networks decide who gets funded, when ride hailing companies determine labour rules. This is the problem of **tech sovereignty.** Without sovereignty, there is no democracy, no market, no self-determination.

Supporting innovation on captured infrastructure is a waste of money. Innovation can only rarely displace infrastructural power: an electric car will not remove a tollbooth on someone else's road. **The engine of the tech industry is not innovation, it is power**. Nothing in digital technology makes sense, except in the light of power. In Silicon Valley, people don't talk about research, they talk about moats, about how to use infrastructure to create power through choke-points. Talking about innovation is just a nice trick to keep the politicians out.

Whenever we bring innovation policy to a geopolitical power struggle, we lose. In order to win for European tech sovereignty, to give European companies the option to compete, to strengthen and promote our democratic society we need to deploy digital infrastructures that are architected to support democratic governance and sovereignty.

What does this mean for social media?

A core concern in digital architecture — just as in institution design — is determining who has **authority** over which action or information. To give an example: if Alice is reading a post from Bob on Twitter, she *has* to trust Twitter that they correctly authenticated Bob, that they verified Bob's identity to some degree, and that they didn't alter the content of Bob's message. In this kind of architecture, Twitter is the authority that determines all of these aspects and more.

This creates **authority monoliths**: systems that are not just choke-holds over one area of responsibility but many, like identity, content, recommendation, advertising, speech rights, social graphs, appeals, or privacy. This creates systems that have extremely concentrated power because instead of having checks and balances between areas of responsibility they become mutually-reinforcing.

The Fediverse approach is to create smaller monoliths. The social system is separated into smaller instances, each of which has authority for its users but that are also connected so that messages can be seen between instances. This is an important step forward because it provides the option to exit an instance without exiting the network. However, it does introduce the challenge in picking the right instance, instances can fail (e.g. by running out of money or being operated by a small dictator), and if one monolith becomes larger than the others (e.g. if Google starts giving all Gmail users a free Fediverse account) then the network will lack the institutional capacity to resist it as the only balancing function is instance size. At the risk of scaring the computer people with technical jargon, given the complexity and power structure of social media, this kind of federation is insufficiently polycentric.

An alternative approach is to architect protocols around what is known technically as **self-certifying data**. Self-certifying data is data that has the ability to prove its own authenticity without resorting to an external authority. Anyone can verify that it is authentic. If Bob asks Eve to deliver a message to Alice, Alice can know with certainty that the message was indeed from Bob even if she doesn't trust Eve at all.

Removing the reliance on central authority opens the door to creating a rich network of institutional arrangements that operate independently from one another, with a separation of powers and responsibilities, just as you would expect from a democratic system. Because the data is verifiable without an external authority, it becomes possible to have separate systems for identity or storing personal data. Recommendation feeds can come from completely arbitrary sources with assurance they aren't lying about the content. And diverse moderation systems can coexist, grounded in the same trustworthy data rather than a single centre of control. This creates democratic affordances, checks and balances, and makes regulation both simpler and more enforceable, while reducing the need for it through the system's increased self-governance — delivering a *much* better user experience.

Eurosky builds on AT (Authenticated Transfer) because **this is the approach taken by the AT Protocol**. It does not exist in opposition to the Fediverse approach (we know from experience how protocol wars are a waste of time), in fact the higher flexibility of self-certifying data protocols means that it is possible to support Fediverse-style governance and experience on top of the AT Protocol, while the reverse is not true. As it nears 40 million users, Bluesky has shown that it is possible to build highly attractive user experiences with this approach, and the new generation of social applications currently being developed on this open protocol shows how much more is possible.

All of these nice architectural properties cannot however be produced if we end up having only one operator running all of those different institutional components. **That is why Eurosky is working to operate public social media infrastructure: to create guarantees that the AT network will remain open and operate in the public interest.**